CSE 3205 Computer Networks
Network Models

The OSI reference model organizes network functions into seven layers: physical, data link, network, transport, session, presentation, and application. The TCP/IP model used on the Internet has four layers: link, internet, transport, and application.
Each layer adds its own header to the data it receives from the layer above, a process called encapsulation. The unit of data is called a frame at the data link layer, a packet at the network layer, and a segment at the transport layer.

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Data Link Layer

The data link layer provides node-to-node delivery across a single link. Error detection commonly uses a cyclic redundancy check, in which the sender appends a remainder computed by polynomial division and the receiver recomputes it. Ethernet uses carrier sense multiple access with collision detection. A station listens before transmitting, and if a collision is detected it stops and waits a random backoff time. Modern switched networks are full duplex, so collisions no longer occur.

A switch learns MAC addresses by reading the source address of incoming frames and storing the port in its forwarding table.

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IP Addressing

Internet Protocol version 4 uses 32-bit addresses, usually written in dotted decimal notation. Classless Inter-Domain Routing, or CIDR, writes a prefix length after the address, such as 192.168.1.0/24, which leaves eight bits for hosts.

Because IPv4 addresses are scarce, Network Address Translation lets many devices in a private network share a single public address. The router rewrites source addresses and tracks connections in a translation table.

IPv6 uses 128-bit addresses, which removes the shortage of addresses and simplifies the packet header.

Routers forward packets using a forwarding table built by routing protocols. In a distance vector protocol such as RIP, each router shares its table with its neighbors and applies the Bellman-Ford algorithm. A known weakness is the count to infinity problem, which slows convergence after a failure.

In a link state protocol such as OSPF, every router floods information about its links and then computes shortest paths with Dijkstra's algorithm over the full topology. Between autonomous systems the Internet uses the Border Gateway Protocol, a path vector protocol that makes decisions based on policy.

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Transport Layer: UDP and TCP

The transport layer provides process-to-process communication using port numbers. UDP is connectionless and offers no delivery guarantee, so it has very low overhead and suits streaming and DNS queries.
TCP is connection-oriented and provides reliable, in-order delivery. A connection is established with a three-way handshake: the client sends SYN, the server replies with SYN-ACK, and the client answers with ACK. Sequence numbers and acknowledgements allow lost segments to be detected and retransmitted.

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TCP Congestion Control

TCP adjusts its sending rate with a congestion window. During slow start the window grows exponentially, doubling every round trip, until it reaches a threshold.
Congestion avoidance then increases it by about one segment per round trip, which is called additive increase.
When three duplicate acknowledgements arrive, TCP performs fast retransmit and halves the window, known as multiplicative decrease. A timeout is treated as a more severe signal and resets the window to one segment.
Flow control is different: the receiver advertises a window so that the sender does not overflow the receiver's buffer.

The Domain Name System translates human-readable names into IP addresses using a hierarchy of root, top-level domain, and authoritative name servers. Resolvers cache answers according to a time-to-live value.

HTTP is a stateless request-response protocol. Cookies let servers keep state across requests. HTTP/2 multiplexes several streams over a single connection, and HTTP/3 runs over QUIC, which is built on UDP.

DHCP automatically assigns an IP address and other configuration to hosts when they join a network.